

Q.1)

a) Derive performance equation for mixed flow reactor.

→ performance equation for mixed flow reactor.

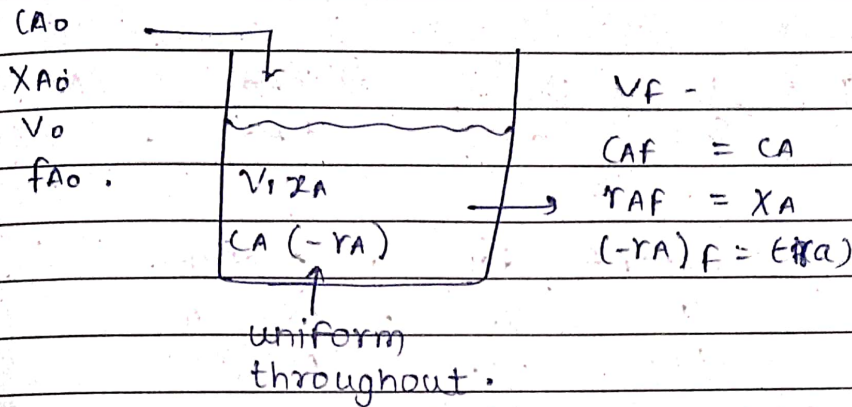


fig. mixed flow reactor.

- Material balance for component A in mixed flow reactor can be written as,

$$\text{Input} = \text{output} + \text{Disappearance by } r_A + \text{Accumulation} \quad (1)$$

As the composition is uniform throughout, eqn (1) reduces to

$$\text{Input} = \text{output} + \text{Disappearance by } r_A \quad (2)$$

Here,

$$\text{Input of A} = f_{A0} (1 - x_{A0}) = f_{A0} \quad (3)$$

(moles/time)

$$\text{output of A} = f_A = f_{A0} (1 - x_A) \quad (4)$$

(moles / time)

$$\text{Disappearance of A} = (-r_A) V \quad (5)$$

by r_A
(moles / time)

putting values of eqn (3), (4) & (5) in eqn (2) we get

$$F_{A0} \cdot X_A = (-r_A) V \quad \text{--- (6)}$$

Rearranging eqn (6).

$$\frac{V}{F_{A0}} = \frac{X_A}{(-r_A)} = \frac{I}{C_{A0}} = \frac{\Delta X_A}{-r_A}$$

$$I = \frac{1}{S} = \frac{V}{V_0} = \frac{V \cdot C_{A0}}{F_{A0}} = \frac{C_{A0} \cdot X_A}{-r_A} \quad \text{--- (7)}$$

for constant density system

$$\frac{V}{F_{A0}} = \frac{X_A}{-r_A} = \frac{C_{A0} - C_A}{C_{A0}(-r_A)}$$

$$I = \frac{C_{A0} \cdot X_A}{-r_A}$$

$$\therefore \boxed{I = \frac{C_{A0} - C_A}{-r_A}} \quad \text{--- (8)}$$

eqn (8) is the performance eqn for MFR.

Q.2) Derive performance equation for Batch reactor

→

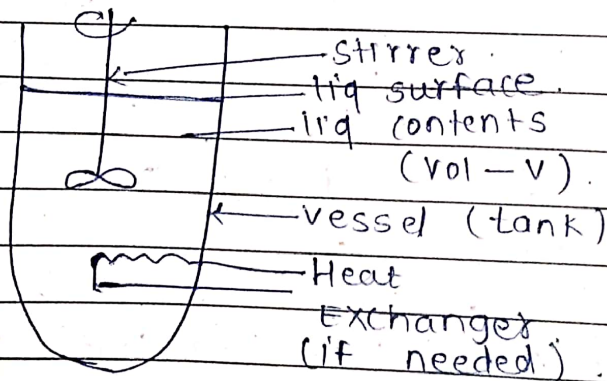


fig. Batch reactor.

In a batch reactor, since the composition is uniform throughout any instant of time we may make accounting about whole react

material balance with respect to reactant A can be written as :

$$\text{Input} = \text{output} + \text{disappearance} + \text{accumulation} \quad (1)$$

As this is a batch operation no fluid enters or leaves the reactor during entire course of ~~takes~~ place reaction.

Therefore,

$$\text{Input} = \text{output} = 0.$$

Hence, we rewrite eqn (1) as,

$$\left[\begin{array}{l} \text{Rate of loss of reactant A} \\ \text{within reactor due to} \\ \text{chemical reaction.} \end{array} \right] = - \left[\begin{array}{l} \text{Rate of Accumulation of reactant} \\ \text{A within the reactor} \end{array} \right]$$

$$\begin{array}{l} \text{Disappearance of A} \\ \text{by reaction} \\ \text{(moles / time)} \end{array} = \left[\frac{\text{moles of reacting}}{(\text{time})(\text{Vol of the fluid})} \right] (\text{Vol of fluid}).$$

$$= (-r_A) V \quad (3)$$

$$\begin{array}{l} \text{Accumulation of A} \\ \text{(moles / time)} \end{array} = \frac{dN_A}{dt} = \frac{d[N_{A0}(1-x_A)]}{dt}$$

$$= - \frac{N_{A0} \cdot dx_A}{dt} \quad (4)$$

eqn (2) can be written as,

$$(-r_A) V = N_{A0} \cdot \frac{dx_A}{dt} \quad (5)$$

Rearranging and integrating eqn (5) to evaluate t we get,

$$t = N_{A0} \cdot \int_0^{x_A} \frac{dx_A}{(-r_A) \cdot V} \quad (6)$$

eqn (6) describes the performance eqn of Batch constant density system.

Q.3)4

with the help of suitable example explain auto-catalytic reactions

- When a material reacts away by any n^{th} order rate in batch reactor, its rate of disappearance is rapid at the start when conc of reactant is high.
- This rate then slows progressively as reactant is consumed

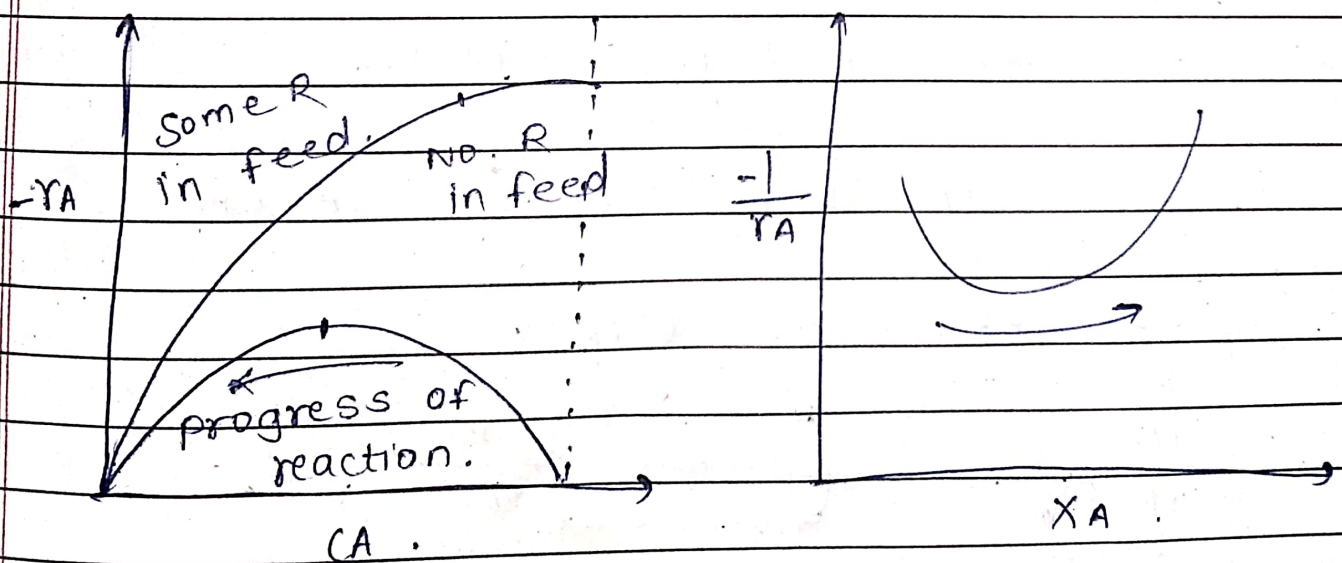
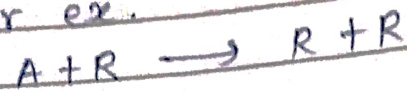


fig: Typical rate-const. curve for autocatalytic reactions.

for ex.



$$-r_A = k C_A^a C_R^r$$

for autocatalytic reactions mixed flow is more efficient at low conversions, plug flow is more efficient at high conversions.

Q.5) Derive performance equation of plug flow reactor.

→

In a plug flow reactor the composition of the fluid varies from point to point along a flow path:

∴ The material balance is

input = output + disappearance by + accumulation
 $\downarrow$ $\downarrow$
 r_A $\downarrow$
 ≤ 0

$$\text{input} = F_A$$

$$\text{output} = F_A + dF_A$$

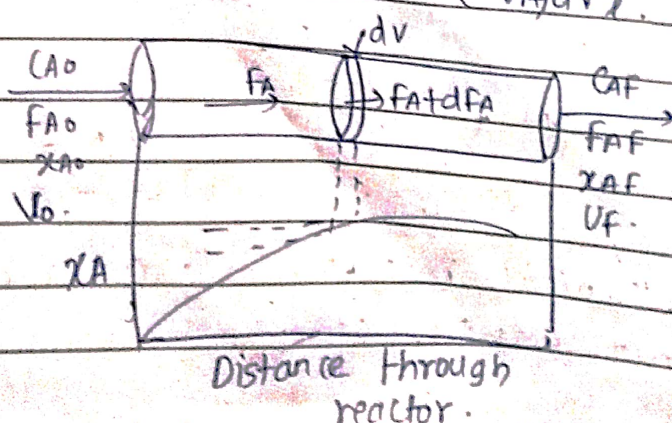
$$\text{disappearance of } A = (-r_A)dv.$$

by reaction

$$\therefore F_A = (F_A + dF_A) + (-r_A)dv.$$

$$\therefore dF_A = d(F_{A0}(1-x_A)) = -F_{A0} dx_A.$$

$$\therefore F_{A0} dx_A = (-r_A)dv.$$



This then eqⁿ which accounts for A in the differential section of reactor of volume dv .
for the reactor as a whole expression must be integrated.

$$\therefore \int_0^V \frac{dv}{f_{A0}} = \int_0^{x_A} \frac{dx_A}{-r_A}$$

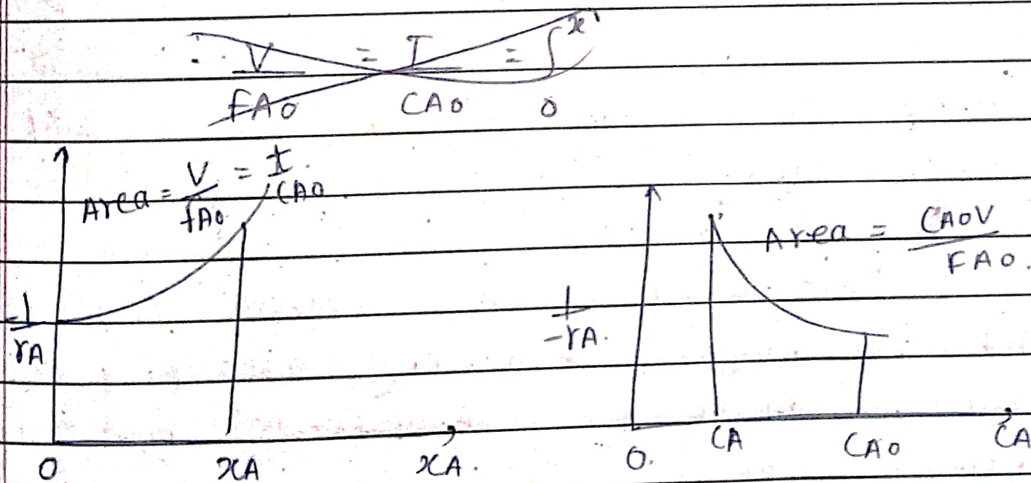
$$\text{Thus } \frac{V}{f_{A0}} = \frac{I}{C_{A0}} = \int_0^{x_{Af}} \frac{dx_A}{-r_A}$$

$$\frac{V}{f_{A0}} = \int_{x_{Ai}}^{x_{Af}} \frac{dx_A}{-r_A}$$

$$\therefore I = C_{A0} \int_{x_{Ai}}^{x_{Af}} \frac{dx_A}{-r_A}$$

For constant density system.

$$x_A = \frac{C_{A0} - C_A}{C_{A0}} \quad \text{and} \quad dx_A = \frac{-dC_A}{C_{A0}}$$



General Case.

fig: Graphical representation of performance equation of PFR.

Q.6) Explain the term space time and space velocity and state ~~its~~ their units.

→ space time: It is time required to process one reactor volume of feed measured at specific condition.

$$\tau = \frac{V}{v_0} \quad \begin{array}{l} V \rightarrow \text{Reactor volume} \\ v_0 \rightarrow \text{volumetric flow rate} \end{array}$$

Its unit is time

space Velocity: It is the number of reactor volumes of feed at specified conditions which can be treated in unit time.

$$S = \frac{1}{\tau}$$

- its unit is 'time⁻¹'

Q.7) Explain the term holding time and space time in detail.

→ holding time → It is also called as residence time which is nothing but the actual average time a fluid element spends inside the reactor.

$$t_h = \frac{V}{v}$$

holding time.

$V \rightarrow$ Reactor volume.
 $v \rightarrow$ volumetric flow rate.

- In ideal condn $t_h = \tau$
- In Real condn $t_h \neq \tau$

Q8) write advantages, disadvantages and applⁿ of
i) Batch reactor ii) mixed flow reactor
iii) plug flow reactor.

① Batch Reactor: A closed system where ~~so~~ reactants are loaded, reaction occurs, and products are removed after the reaction is complete. No flow in or out during the reaction.

• Advantages

- simple to operate and design
- Good for small scale production and lab testing
- flexible: suitable for producing different type products in the same equipment.
- High conversion per batch possible for irreversible reactions.

• Disadvantages

- Not suitable for large scale continuous prodⁿ
- Labor & time intensive due to loading / unloading
- poor temp & press control in large vol
- Hard to automate & scale.

time. • Applⁿ.

- pharmaceutical production
- fine chemicals and specialty chemicals.
- food processing
- pilot plants and research labs.

② mixed flow reactor (CSTR): continuous input & output with complete mixing assumed. The output concentration is the same as inside the reactor.

- Advantages.
 - Easy to control temp due to uniform mixing
 - Continuous operation - suitable for large scale processes
 - steady state operation possible.
 - ideal for liquid - phase rxn with slow kinetics

- Disadvantages.
 - Lower conversion for given volume: compare to PFR
 - Larger reactor volume needed for high conversion
 - mixing may not be perfect in real systems.

- Applications.
 - Wastewater treatment
 - Homogeneous liq - phase rxn.
 - polymerization & fermentation processes.
 - Rxn requiring constant environmental cond^{ns}

③ PFR : Reactants flow through a cylindrical pipe with no back-mixing. composition and rxn progress vary along the length of the reactor.

- Advantages
 - higher conversion per reactor volume for many reaction
 - efficient for fast reactions
 - Good for large - scale, continuous processing
 - predictable performance with well-defined residence time

- Disadvantages
 - Difficult temp control
 - complex design and maintenance
 - not ideal for rxn requiring mixing or multiple

Application

petrochemical & refinery industries

Gas-phase and high temp reactions

Large scale chemical prodⁿ

Catalytic zens in tubular reactors

g) write a short note on reactors of different types in series.

→ If reactors of different types are put in series, such as a mixed flow reactor followed by a plug flow reactor which in turns is followed by another mixed flow reactor, we may write for the three reactors

$$\frac{V_1}{f_0} = \frac{x_1 - x_0}{(-r_1)}$$

$$\frac{V_2}{f_2} = \int_{x_1}^{x_2} \frac{dx}{-r}$$

$$\frac{V_3}{f_3} = \frac{x_3 - x_2}{(-r_3)}$$

This allows us to predict the overall conversions for such systems, or conversions at intermediate points between the individual reactors.

• Best arrangement of a set of ideal reactors:

1) for a reaction whose rate-concentration curve rises monotonically the reactors should be connected in series. they should be ordered so as to keep the concentration of reactant as high as possible.

2) for reactions where the rate-concentrations curve passes through maximum or minimum the arrangement of units depends on the actual shape of curve, the conversion level desired, &

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 9) whatever may be the kinetics and the reactor system, an examination of the $1/(-r_A)$ vs C_A curve is a good way to find the best arrangement of units.

Q.10) Explain series, parallel and independent reaction with suitable example.

→

① Series reaction.

In series reaction, the product of one reaction becomes the reactant for the next.

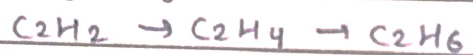
General form : $A \rightarrow B \rightarrow C$

A - initial reactant

B - intermediate

C - final product.

example: Hydrogenation of ethane.

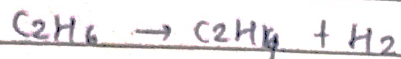


② Parallel reaction

In a parallel reaction, a single reactant forms multiple products through different pathways simultaneously.

General form $A \rightarrow B$
 $A \rightarrow C$

example: Thermal decomposition of ethane

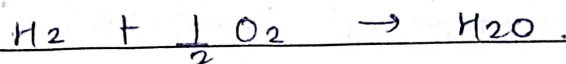
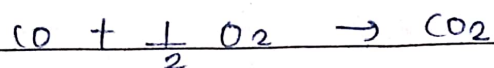


③ Independent Reaction.

Independent reactions involve different reactants and products proceed independently of each other.

General form : $A \rightarrow B$
 $C \rightarrow D$

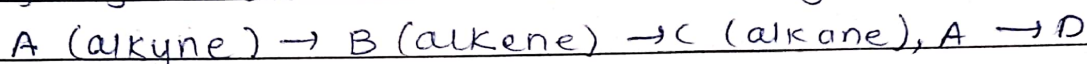
example :



④ complex Reⁿ.

- combinations of series and parallel reactions.

ex. hydrogenation of unsaturated hydrocarbon



⑤ Reversible Reaction

products can revert to reactants establishing equilibrium.

General form:

